

CA1 – Task3: Dashboard improvement

Mani Kermani – Alireza Rastegar - Fereshte Kambarani

Audit:

The original dashboard provided visualization of stock market data but lacked advanced analytical capabilities, focused storytelling, and effective visual hierarchy. Although the dashboard contained numerous visual elements, important financial metrics and market patterns were difficult to identify quickly due to visual clutter and the overuse of complex charts such as radar and violin plots.

Additionally, the dashboard was not sufficiently time-intelligent and provided limited support for dynamic time-based analysis and interactive exploration. While raw statistical patterns were displayed, the dashboard lacked advanced financial indicators, summarized KPI metrics, and insight-driven components that could support faster interpretation, trend analysis, momentum evaluation, and volatility assessment.

Solution:

To address the limitations of the original dashboard, the redesigned solution focused on improving analytical depth, interactivity, visual clarity, and financial storytelling. Several technical and visual enhancements were implemented using Microsoft Power BI and DAX measures.

First, a dedicated date table and time-intelligent measures were introduced to support dynamic time-based analysis. Interactive slicers were added to allow users to filter and explore stock behavior across different time periods more efficiently.

The date table was implemented using the following DAX expression.

```
date_table =  
CALENDAR(  
    MIN('khodro_final_daily'[date]),  
    MAX('khodro_final_daily'[date])  
)
```

To improve analytical capabilities, advanced financial indicators were implemented, including 10-day and 30-day moving averages (MA10 and MA30), RSI threshold analysis, and rolling volatility measures. These indicators enable users to identify bullish and bearish trends, overbought and oversold conditions, and periods of increased market uncertainty. The DAX codes are shown below.

```
MA10 =  
CALCULATE(  
    AVERAGE(khodro_final_daily[close]),  
    DATESINPERIOD(  
        date_table[Date],  
        MAX(date_table[Date]),
```

```

        -10,
        DAY
    )
)
MA30 =
CALCULATE(
    AVERAGE(khodro_final_daily[close]),
    DATESINPERIOD(
        date_table[Date],
        MAX(date_table[Date]),
        -30,
        DAY
    )
)

```

The RSI indicator was implemented using the following DAX measures:

```

price_change =
MAX(khodro_final_daily[close]) -
MAX(khodro_final_daily[open])

```

```

gain =
IF(
    [price_change] > 0,
    [price_change],
    0
)

```

```

loss =
IF(
    [price_change] < 0,
    ABS([price_change]),
    0
)

```

```

avg_gain_14 =
CALCULATE(
    AVERAGEX(
        VALUES(date_table[Date]),
        [gain]
    ),
    DATESINPERIOD(
        date_table[Date],
        MAX(date_table[Date]),
        -14,
        DAY
    )
)

```

```

    )
)

avg_loss_14 =
CALCULATE(
    AVERAGEX(
        VALUES(date_table[Date]),
        [loss]
    ),
    DATESINPERIOD(
        date_table[Date],
        MAX(date_table[Date]),
        -14,
        DAY
    )
)

```

```

RS =
DIVIDE(
    [avg_gain_14],
    [avg_loss_14]
)

```

```

RSI =
100 -
DIVIDE(
    100,
    1 + [RS]
)

```

And for rolling volatility:

```

daily_return =
DIVIDE(
    MAX(khodro_final_daily[close]) -
    MAX(khodro_final_daily[open]),
    MAX(khodro_final_daily[open])
)

```

```

rolling_volatility_30 =
CALCULATE(
    STDEVX.P(
        VALUES(date_table[Date]),
        [daily_return]
    )
)

```

```

    ),
    DATESINPERIOD(
        date_table[Date],
        MAX(date_table[Date]),
        -30,
        DAY
    )
)

```

KPI cards were added to highlight important financial metrics such as latest close price, RSI values, trading activity, and market volatility. This improved visual hierarchy and reduced cognitive load by making key information immediately visible.

Custom tooltip pages and dynamic insight panels were also implemented to improve dashboard storytelling and reduce visual clutter. These additions allow users to access contextual information interactively while maintaining a cleaner main dashboard layout.

Several UX/UI principles guided the redesign process, including consistent color usage, simplified visual structure, improved spacing and alignment, and reduced dependence on visually complex charts. The redesigned dashboard provides a more focused, interactive, and insight-driven environment for financial analysis and decision-making.

Note: Some DAX codes for some measures are not in the report but you can see in our .pbix file.

Insight discovery:

The redesigned dashboard enables several analytical insights that were difficult or impossible to identify in the original version. By integrating advanced financial indicators, KPI summaries, and interactive exploration tools, users can now analyze stock behavior more effectively and identify meaningful market patterns.

The implementation of **MA10 and MA30** moving averages allows users to detect **bullish** and **bearish** momentum shifts through moving average crossover analysis.

In addition, **rolling volatility** analysis provides insight into periods of increased **market instability** and financial risk. Interactive slicers and time-intelligent measures allow users to dynamically explore market behavior across custom time intervals, improving temporal analysis and trend comparison.

RSI threshold visualization enables the identification of **overbought** and **oversold** market conditions and improves momentum analysis.

The addition of KPI cards, tooltips, and insight panels also improves rapid interpretation and decision-making by summarizing important financial conditions in a clear and accessible format.